

Curriculum Vitae

Dr Michael Lydeamore
Department of Econometrics and Business Statistics
Monash University
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APPOINTMENTS HELD

Senior Lecturer - Econometrics and Business Statistics	2021-
Monash University	
COVID-19 Modelling & Forecasting Lead	2020
Manager, Analytics	
Department of Health and Human Services, Victoria	
Postdoctoral Research Fellow	2019-2021
Monash University	
Honorary Team Member	2019-
SaferCare Victoria	
Honorary Research Fellow	2019-
Alfred Health	
Research Fellow	2018-2019
The Kirby Institute, University of New South Wales	

EDUCATION

Doctor of Philosophy - Applied Mathematics	2015-2019
The University of Melbourne	
Thesis title: Mechanistic and statistical models of skin disease transmission	
Masters of Philosophy - Applied Mathematics	2013-2014
The University of Adelaide	
Thesis title: Approximations of stochastic household models for comparing antiviral allocation schemes	
Awarded the Applied Probability Trust prize for the highest mark with a project in Applied Mathematics and Statistics	
Bachelor of Mathematical Sciences	2010-2013
The University of Adelaide	
Graduated with a double major in applied and pure mathematics	

PUBLICATIONS

PREPRINTS

1. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *quollr: An R Package for Visualizing 2-D Models from Nonlinear Dimension Reductions in High-Dimensional Space* (arXiv:2512.18166). arXiv. <https://doi.org/10.48550/arXiv.2512.18166>
2. Gamage, J. P., Cook, D., Harrison, P., **Lydeamore, M.**, & Talagala, T. S. (2025). *cardinalR: Generating Interesting High-Dimensional Data Structures* (arXiv:2512.18172). arXiv. <https://doi.org/10.48550/arXiv.2512.18172>

3. Gamage, J. P., Cook, D., Harrison, P., & **Lydeamore, M.** (2025). *Stop Lying to Me: New Visual Tools to Choose the Most Honest Nonlinear Dimension Reduction* (arXiv:2506.22051). arXiv. <https://doi.org/10.48550/arXiv.2506.22051>
4. Miller, C. M., **Lydeamore, M. J.**, Waddle, A. W., Berger, L., Skerratt, L. F., Flegg, J. A., & Campbell, P. T. (2025). *Mathematical modelling of chytridiomycosis transmission in frogs* (arXiv:2503.06846). arXiv. <https://doi.org/10.48550/arXiv.2503.06846>
5. Conway, E., Walker, C., **Lydeamore, M.**, Golding, N., Ryan, G., Mavec, D., Oates, J., Kabashima, G., Price, D. J., Shearer, F., Cromer, D., Davenport, M. P., McCaw, J., Eriksson, E. M., Hodgkin, P. D., Wu, L., Le, T. P., Baker, C. M., Mueller, I., & McVernon, J. (2024). *Optimal timing of booster doses in a highly vaccinated population with minimal natural exposure to COVID-19*. medRxiv. <https://doi.org/10.1101/2024.05.14.24307386>

PEER-REVIEWED ARTICLES

1. **Lydeamore, M.**, Zachreson, C., Conway, E., Shearer, F. M., Baker, C. M., Ross, J. V., Miller, J., McCaw, J. M., Geard, N. L., McVernon, J., & Price, D. J. (2026). Border quarantine, vaccination and public health measures to mitigate the impact of COVID-19 importations in Australia: a modelling study. *Journal of The Royal Society Interface*, 23(235), 20250144. <https://doi.org/10.1098/rsif.2025.0144>
2. Tierney, N., Saraswati, C., Babu, A., **Lydeamore, M.**, & Golding, N. (2026). conmat: generate synthetic contact matrices for a given age-stratified population. *Journal of Open Source Software*, 11(118), 8326. <https://doi.org/10.21105/joss.08326>
3. Nguyen, Q., **Lydeamore, M. J.**, Dessauer, A., Snell, G. I., Westall, G. P., Peleg, A. Y., & Gardiner, B. J. (2026). Evaluation of Global Immune Biomarkers to Predict Infections in Lung Transplant Recipients. *Clinical Transplantation*, 40(8), e70654. <https://doi.org/10.1111/ctr.70654>
4. Le, T. P., Conway, E., Akpan, E., Abell, I. R., Abraham, P., Baker, C. M., Campbell, P. T., Cromer, D., **Lydeamore, M. J.**, McDonough, Y., Mueller, I., Ryan, G., Walker, C., Wang, Y., Carvalho, N., & McVernon, J. (2025). Health impact and cost-effectiveness of COVID-19 booster vaccination strategies in the early post-Omicron era: a dynamic modelling study. *BMJ Global Health*, 10(9). <https://doi.org/10.1136/bmjgh-2024-016096>
5. Hao, T., Ryan, G. E., **Lydeamore, M. J.**, Cromer, D., Wood, J. G., McVernon, J., McCaw, J. M., Shearer, F. M., & Golding, N. (2025). Predicting immune protection against outcomes of infectious disease from population-level effectiveness data with application to COVID-19. *Vaccine*, 55, 126987. <https://doi.org/10.1016/j.vaccine.2025.126987>
6. **Lydeamore, M. J.**, Wu, D., Donker, T., Gorrie, C., Higgs, C. K., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Changes in isolation guidelines for CPE patients results in only mild reduction in required hospital beds. *Infection, Disease & Health*. <https://doi.org/10.1016/j.idh.2024.10.004>
7. **Lydeamore, M. J.**, Donker, T., Wu, D., Gorrie, C., Turner, A., Easton, M., Hennessy, D., Geard, N., Howden, B. P., Cooper, B. S., Wilson, A., Peleg, A. Y., & Stewardson, A. J. (2024). Carbapenemase-producing enterobacterales colonisation status does not lead to more frequent admissions: a linked patient study. *Antimicrobial Resistance & Infection Control*, 13(1), 82. <https://doi.org/10.1186/s13756-024-01437-x>
8. Shearer, F. M., Edwards, L., Kirk, M., Eales, O., Golding, N., Hassall, J., Liu, B., **Lydeamore, M.**, Miller, C., Moss, R., Price, D. J., Ryan, G. E., Sheena, S., Tobin, R., Ward, K., Kaldor, J., Cheng, A. C., Wood, J., & McCaw, J. M. (2024). Opportunities to strengthen respiratory virus surveillance systems in Australia: lessons learned from the COVID-19 response. *Communicable Diseases Intelligence*, 48. <https://doi.org/10.33321/cdi.2024.48.47>

9. Shearer, F. M., McCaw, J. M., Ryan, G. E., Hao, T., Tierney, N. J., **Lydeamore, M. J.**, Wu, L., Ward, K., Ellis, S., Wood, J., McVernon, J., & Golding, N. (2024). Estimating the impact of test–trace–isolate–quarantine systems on SARS-CoV-2 transmission in Australia. *Epidemics*, 47, 100764. <https://doi.org/10.1016/j.epidem.2024.100764>
10. Le, T. P., Abell, I., Conway, E., Campbell, P. T., Hogan, A. B., **Lydeamore, M. J.**, McVernon, J., Mueller, I., Walker, C. R., & Baker, C. M. (2024). Modelling the impact of hybrid immunity on future COVID-19 epidemic waves. *BMC Infectious Diseases*, 24(1), 407. <https://doi.org/10.1186/s12879-024-09282-4>
11. Conway, E., Walker, C. R., Baker, C., **Lydeamore, M. J.**, Ryan, G. E., Campbell, T., Miller, J. C., Rebuli, N., Yeung, M., Kabashima, G., Geard, N., Wood, J., McCaw, J. M., McVernon, J., Golding, N., Price, D. J., & Shearer, F. M. (2023). COVID-19 vaccine coverage targets to inform reopening plans in a low incidence setting. *Proceedings of the Royal Society B: Biological Sciences*, 290(2005), 20231437. <https://doi.org/10.1098/rspb.2023.1437>
12. Mitchell, B. G., Stewardson, A. J., Kerr, L., Ferguson, J. K., Curtis, S., Busija, L., **Lydeamore, M. J.**, Graham, K., & Russo, P. L. (2023). The incidence of nosocomial bloodstream infection and urinary tract infection in Australian hospitals before and during the COVID-19 pandemic: an interrupted time series study. *Antimicrobial Resistance & Infection Control*, 12(1), 61. <https://doi.org/10.1186/s13756-023-01268-2>
13. Zachreson, C., Shearer, F. M., Price, D. J., **Lydeamore, M. J.**, McVernon, J., McCaw, J., & Geard, N. (2022). COVID-19 in low-tolerance border quarantine systems: Impact of the Delta variant of SARS-CoV-2. *Science Advances*, 8(14), eabm3624. <https://doi.org/10.1126/sciadv.abm3624>
14. **Lydeamore, M. J.**, Mitchell, B. G., Bucknall, T., Cheng, A. C., Russo, P. L., & Stewardson, A. J. (2022). Burden of five healthcare associated infections in Australia. *Antimicrobial Resistance & Infection Control*, 11(1), 69. <https://doi.org/10.1186/s13756-022-01109-8>
15. Trauer, J. M., **Lydeamore, M. J.**, Dalton, G. W., Pilcher, D., Meehan, M. T., McBryde, E. S., Cheng, A. C., Sutton, B., & Ragonnet, R. (2021). Understanding how Victoria, Australia gained control of its second COVID-19 wave. *Nature Communications*, 12(1), 6266. <https://doi.org/10.1038/s41467-021-26558-4>
16. McMahon, J. H., **Lydeamore, M. J.**, & Stewardson, A. J. (2021). Bringing evidence from press release to the clinic in the era of COVID-19. *Journal of Antimicrobial Chemotherapy*, 76(3), 547–549. <https://doi.org/10.1093/jac/dkaa506>
17. Zachreson, C., Mitchell, L., **Lydeamore, M. J.**, Rebuli, N., Tomko, M., & Geard, N. (2021). Risk mapping for COVID-19 outbreaks in Australia using mobility data. *Journal of The Royal Society Interface*, 18(174), 20200657. <https://doi.org/10.1098/rsif.2020.0657>
18. Sullivan, S. G., Brotherton, J. M., Lynch, B. M., Cheung, A., **Lydeamore, M.**, Stevenson, M., Firestone, S., Canevari, J., Nguyen, H. N. J., & Carville, K. S. (2021). Population-based analysis of the epidemiological features of COVID-19 epidemics in Victoria, Australia, January 2020–March 2021, and their suppression through comprehensive control strategies. *Lancet Regional Health - Western Pacific*, 17.
19. **Lydeamore, M. J.** (2021). Mathematical models to support Victoria’s COVID-19 response: a blunt instrument to a complex problem. *Journal of the Australian Mathematical Society*. <https://austms.org.au/wp-content/uploads/2021/07/Lydeamore.pdf>
20. **Lydeamore, M. J.**, Campbell, P. T., Price, D. J., Wu, Y., Marcato, A. J., Cuningham, W., Carapetis, J. R., Andrews, R. M., McDonald, M. I., McVernon, J., Tong, S. Y. C., & McCaw, J. M. (2020). Estimation of the force of infection and infectious period of skin sores in remote Australian communities using interval-censored data. *PLOS Computational Biology*, 16(10), e1007838. <https://doi.org/10.1371/journal.pcbi.1007838>

FUNDING

Almost all of this funding was obtained in partnership with large, collaborative teams.

SPARKLE	2024
Department of Foreign Affairs and Trade	
Awarded value: \$5,000,000	
SHIELD: Surveillance of Healthcare-associated Infections for Effective Local Data	2025
NHMRC	
Awarded value: \$2,000,000	
Modelling Work in Low and Middle Income Counties in the Western Pacific Region for the COVID-19 SAGE Working Group	2024
World Health Organisation	
Awarded value: \$8,157	
Modelling To Support Australia's Resilience to and Preparedness for Omicron And Future Sars-Cov-2 Variants	2024
Commonwealth Government of Australia	
Awarded value: \$80,059	
Contractual Partner to Conduct Research on Healthcare-Associated Infection Rates in Phillippine Hospitals	2023
World Health Organisation Phillippines	
Awarded value: \$72,078	
Centre of Western Public Health Unit	2023
Contract Research	
Awarded value: \$25,000	
Extending and comparing methods for projecting social contact matrices	2022
SPECTRUM/SPARK Seed Funding	
Awarded value: \$19,438	
Quantifying longitudinal relationships between community mobility and COVID-19 case incidence in west metropolitan Melbourne	2022
SPECTRUM/SPARK Seed Funding	
Awarded value: \$19,896	
Modelling to support Australia's national plan for COVID-19	2021
Commonwealth Government of Australia	
Awarded value: \$147,536	
Modelling to support Australia's transition to 'COVID-normal'	2021
Australian Office of Health Protection	
Awarded value: \$25,620	
CONFERENCE PRESENTATIONS AND CONTRIBUTED TALKS	
Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
International Congress on Teaching Statistics	
Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
useR! Conference	
Gamifying data visualisation: Teaching ggplot2 through competitive code golf	2026
Australia and New Zealand Industrial Applied Mathematics Conference	
Modelling chytridiomycosis transmission in frogs	2025
Australia and New Zealand Industrial Applied Mathematics Conference	
airpurifyr: Open Air Quality Data in R	2024
WOMBAT Annual Meeting: Open the world with open source	

Networks of networks in infectious diseases modelling - Panelist	2024
Infectious Diseases Modelling Conference	
Generating synthetic contact matrices using open-source data	2024
Australia and New Zealand Industrial Applied Mathematics Conference	
Data-Driven Insights into Healthcare Challenges: Two Case Studies - Invited Speaker	2023
Universitätsklinikum Freiburg Seminar Series	
Data-Driven Insights into Healthcare Challenges: Two Case Studies	2023
UNSW Australia Statistics & Data Science Seminar	
Burden of healthcare associated infections in Australia	2022
Australia and New Zealand Industrial Applied Mathematics Conference	
Exponential Random Graph Models and CPE transmission - Invited Speaker	2021
Melbourne Mathematical Biology Seminar Series	
Mathematical modelling for COVID-19 in Victoria, Australia - Invited Speaker	2021
Math for Industry Forum, Vietnam	
The associations between transmission of CPE and ward connectivity: a network analysis	2021
Australasian Society for Infectious Diseases Annual Scientific Meeting	
The burden of healthcare acquired infections in Australian public hospitals	2021
Australasian Society for Infectious Diseases Annual Scientific Meeting	
Decision Making and Mathematical Biology - Victoria's usage of COVID-19 modelling -	2021
Invited Speaker	
Mathematical Biology Special Interest Group Workshop	
Modelling between-household effective contact and the elimination of COVID-19 in Melbourne, Australia	2021
Australia and New Zealand Industrial Applied Mathematics Conference	
Mathematical modelling and Victoria's response to COVID-19 - Invited Speaker	2020
Victorian ANZIAM Branch Meeting	
Australia's experience and the role of modelling in its responses to COVID-19 - Invited	2020
Speaker	
Usher Institute COVID-19 Webinar	
Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data	2019
Australia and New Zealand Industrial Applied Mathematics Conference	
Estimating epidemiological quantities for skin sores in remote Australian communities using interval-censored data	2018
NSW-ACT ANZIAM Branch Meeting	
Coupled models of Group A Streptococcus and Scabies: How likely is eradication?	2018
PRISM International Conference	
Investigating the dynamics of coupled epidemiological transmission models with application to Group A Streptococcus and Scabies	2018
Society for Mathematical Biology Annual Meeting	
Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies	2018
Australia and New Zealand Industrial Applied Mathematics Conference	
Quantifying the age of first infection with skin sores in five remote Australian Aboriginal communities	2017
Lancefield International Symposium on Streptococci and Streptococcal Diseases	

Investigating the dynamics of coupled models with applications to Group A Streptococcus and Scabies	2017
PRISM Annual Knowledge Transfer and Training Conference	
Constructing mathematical models of Group A Streptococcus and Scabies in remote Australian Indigenous communities	2017
- Invited Speaker	
Zeeman Institute Seminar Series	
Developing a model for the transmission and treatment dynamics of scabies infections a high prevalence setting	2017
Melbourne-Manchester Joint Workshop	
Developing a model for the transmission of Group A Streptococcus	2017
Australia and New Zealand Industrial Applied Mathematics Conference	
Determining the age of first infection from incomplete data	2016
Modelling Emerging Infections and Neglected Tropical Diseases Workshop	
Investigating Intervention Intervals for Scabies Infections	2016
Australia and New Zealand Industrial Applied Mathematics Conference	

TEACHING

LECTURER

[ETC5513] Reproducible and Collaborative practices	- Chief Examiner	2024-
[ETC5523] Communicating with Data	- Chief Examiner	2023-
[ETO5513] Reproducible and Collaborative practices	- Chief Examiner	2024-
[ETC5512] Wild-Caught Data		2022-2023
[ETC5521] Exploratory Data Analysis		2022

TUTOR

[MAST10016] Mathematics for Biomedicine	2016-2019
[MAST30001] Stochastic Modelling	2018

TEACHING SERVICE

[M6036] Masters of Health Data Analytics	- Course Management Committee	2024-
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AWARDS

Fellow of the Higher Education Academy	2024
SPECTRUM Annual Meeting - Best Presentation	2024
University of Melbourne Faculty of Engineering and Information Technology Excellence Award in Interdisciplinary Research	2021
IPAA Spirit of Service Awards - Finalist	2021
Engagement Australia Excellence Awards - Outstanding Engagement for Research Impact - Finalist	2021
Top Poster Award - European Congress of Clinical Microbiology & Infectious Diseases	2021
IPAA Victoria Leadership in the Public Sector Awards - Finalist	2021
Best PhD Student Presentation - PRISM² Annual Conference	2017
Applied Probability Trust Prize - for best Applied Mathematics Postgraduate Thesis	2016
Australian Postgraduate Award	2015
Dean's Commendation for Thesis Excellence	2015

ACADEMIC SERVICE & COMMUNITY ENGAGEMENT

Equity Diversity & Social Inclusion Committee	2024-
Department representative	
International Conference on Prevention and Infection Control	2025
Abstract Reviewer	
SPECTRUM Annual Meeting	2024
Organising committee	
SPARK Short Course in Mathematical Diseases Modelling - OUCRU	2023
Facilitator	
International Conference on Prevention and Infection Control	2023
Abstract Reviewer	
Maths in Industry Study Group - NSW Health	2023
Moderator	
WOMBAT Communicating with Data Workshop	2023
Organising Committee	
Australian and New Zealand Industrial and Applied Mathematics Executive Committee	2022-
Treasurer	
Australian and New Zealand Industrial and Applied Mathematics Executive Committee	2021-2022
Early Career Representative	
SPECTRUM-SPARK Early Career Researcher Committee	2022-
Chair	
SPARK Short Course in Mathematical Diseases Modelling - Mahidol University	2022
Facilitator	
Research Tools Workshop in R	2022
Facilitator	
Mathematical Biology Special Interest Group	2019-2022
Treasurer	
ANZIAM 2021 Conference	2020-2021
Organising Committee (Treasurer)	
Computational Biology Research Initiative - The University of Melbourne	2017
Postgraduate Representative	
Computational Biology Postgraduate Sports Group - The University of Melbourne	2017-2018
Secretary	
Maths in Industry Study Group	2017, 2018, 2019
Participant	
The ConocoPhillips Science Experience - The University of Melbourne	2016-2018
Activity Organiser	
Mathematicians in Schools - CSIRO	2016-2018
Project Supervisor/Mentor	
Reviewer	
PLoS Computational Biology; Scientific Reports; Journal of Applied Mathematics; Journal of Statistical Software; Epidemiology & Infection; Mathematics; Medical Journal of Australia; International Tropical Health; Journal of Antimicrobial Resistance; The R Journal	

PROFESSIONAL MEMBERSHIPS

Australian and New Zealand Industrial and Applied Mathematics

2015-

Including the Mathematical Biology Special Interest Group

Australian Mathematics Society

2015-

Statistical Society of Australia

2024-